Three PM Quant Fund

An AI-assisted paper investment platform

Equity momentum, daily pairs trading and crypto

FE5209 investor roadshow
17 October 2026

The investor opportunity

A proposed way to review several quantitative strategies as one accountable fund

Investment problem

- A single trading style can struggle when conditions change
- Separate strategy reports can hide overlapping exposure
- Investors need to understand why capital is at risk

Our fund design

- Three PM mandates with separate virtual accounts
- One Investment Committee and common risk controls
- AI explanations linked to computed strategy evidence

Diversification and better investor understanding are hypotheses to test

Fund governance

Proposed presentation roles with clear responsibility for capital

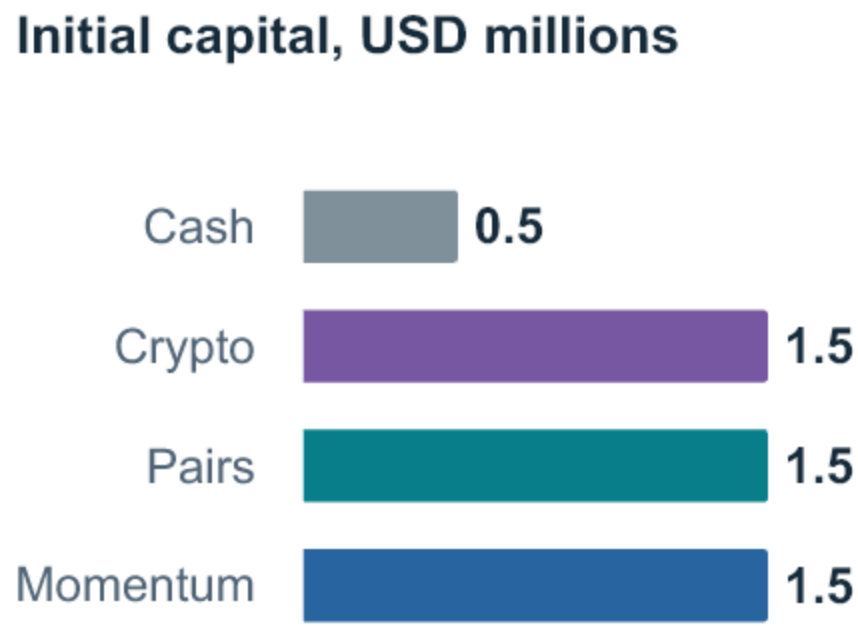


Independent coded risk approval precedes execution

USD 5m paper-capital mandate

Illustrative starting capital, subject to the course convention

Mandate	USD	Share
Momentum PM	1,500,000	30%
Pairs PM	1,500,000	30%
Crypto PM	1,500,000	30%
Central cash	500,000	10%
Total fund	5,000,000	100%



Initial budgets stay fixed. Returns cause each PM’s fund weight to drift.

Current prototype: momentum capital stays in cash until its engine connects.

The three PM mandates

Each PM owns its strategy and an attributable virtual portfolio

PM	Investment approach	Current status
Equity momentum	Teammate stock-selection rule Universe and cadence to confirm	Integration pending
Daily pairs	Teammate Kalman hedge estimate and mean-reverting spread forecast	Adapter connected V/MA research setup
Crypto	Existing crypto signal engine Long exposure or cash	Adapter connected Daily baseline separate

Common PM output: target holdings, timestamps, model version and evidence

The multi-agent AI role

Separate PM contexts, diagnostic tools and evidence-linked explanations

AI representative	Questions it investigates	Investor-facing memo
Momentum PM AI	Rankings, concentration and proposed holdings	Why these stocks? Where is exposure concentrated?
Pairs PM AI	Spread signal, hedge ratio and model health	Why this linked trade? What could invalidate the spread?
Crypto PM AI	Signal qualification, costs and volatility	Why take exposure or hold cash? Which inputs need review?

Planned AI committee chair compares PM memos and records unresolved issues

Current tools and imported replies exist. Automatic model calls remain pending.

The Investment Committee meeting

Illustrative demonstration sequence, to replace with actual saved tool traces

Participant	Review question	Evidence shown
Pairs PM AI	Do both legs preserve the intended hedge?	Signed target legs and hedge-ratio diagnostics
Crypto PM AI	Why does the engine propose exposure or cash?	Signal qualification and local risk verdict
Committee / CIO	Where do PM proposals share fund exposure?	Combined gross exposure and symbol limits
Risk controller	Does the proposal satisfy the approved policy?	Coded approval or rejection with a saved reason

The Board reviews policy and exceptions. The ledger preserves each decision.

The daily pairs decision cycle

Daily frequency is confirmed. Exact signal and fill timing still needs agreement.

Time	Owner	Output
After session t close	Data and Pairs PM	Validated closes and linked target legs
Before execution	Investment Committee / CIO	Mandate policy and combined exposure
Before execution	Independent risk controller	Approval or rejection with reasons
Next eligible session	Proposed paper-fill adapter	Costed fills, subject to borrow and leg handling
After valuation	Accounting and PM reporting	Fund NAV, closing-price CSV and memos

Current integration records intentions and zero fills

Independent risk and fund accounting

The committee works within a declared risk policy

Current coded controls

- 80% fund gross-exposure limit
- 20% gross exposure per symbol
- Linked scaling preserves each pair's hedge
- 25% fund drawdown breaker stays latched

Accounting still to complete

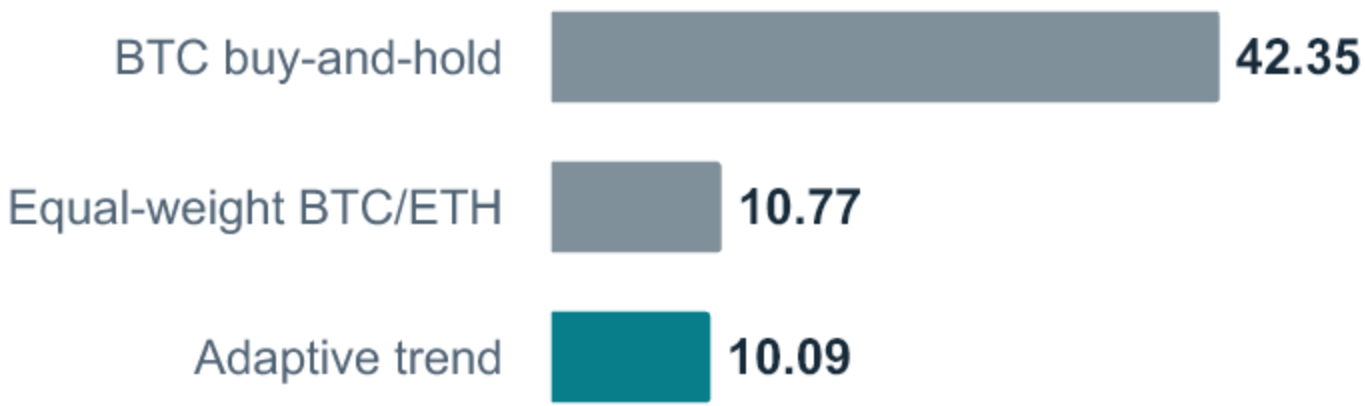
- Confirm holdings before valuing the account
- Add costed fills and short-position financing
- Reconcile PM P&L with consolidated fund NAV
- Record capital transfers separately from return

Fund NAV = sum of PM NAVs + central cash

Historical crypto evidence

Separate daily baseline, 1 January 2024 to 31 July 2026 validation

Total return, percent



Baseline risk

−23.78%

Maximum drawdown

0.32

Sharpe, zero risk-free rate

10 bps turnover cost model. The baseline trails both reported benchmarks.

Historical baseline evidence does not establish integrated-fund or October performance.

Evidence for the investor review

Current local capability and the remaining performance gap

Evidence	Current position
Two-PM local integration	Crypto and daily pairs proposals, diagnostics and risk Recorded intentions with zero fills
Momentum and connected AI	Momentum package and automatic AI calls pending Optional imported replies already supported
Combined fund performance	No costed consolidated return history yet Initial USD 5m account is supplied capital
October evaluation	Course window: 1–23 October 2026 Preserve genuine forward records and label replays

Final scorecard: net return, benchmark, Sharpe, drawdown and trading statistics

The hosted investor demonstration

One small application is enough for the first version

Investor view

- Fund NAV and each PM’s contribution
- Current holdings and capital mandates
- PM explanations with source diagnostics
- Committee decisions and risk verdicts

Proposed operating setup

- One Streamlit dashboard
- Python runner on a daily schedule
- SQLite on persistent storage
- Server-side credentials and a saved run history

Class demo: one saved day, actual inputs, PM proposals, risk decision and receipts

Submission and team accountability

Saturday, 24 October 2026, according to the new course announcement

Required Canvas ZIP

- Editable presentation slides
- Supporting data and closing-price CSV
- Reproducible source code
- Named individual contribution statement

Required registration sheet

- Group name and every member’s details
- Out-of-sample strategy performance
- Class-accessible video link, if prepared

Optional pitch: maximum 60 seconds. A narrated PowerPoint is sufficient.

Role owners: Momentum PM, Pairs PM, Crypto PM, integration/risk and reporting

The investment case

A paper pilot whose decisions classmates can inspect

Three specialist PM mandates

An Investment Committee accountable to a human Board

AI explanations grounded in strategy diagnostics

Our ask: assess the pilot using the final costed out-of-sample evidence